

Context4Change: A Participatory Framework for Context-Driven Behavioral Interventions in Office Well-being

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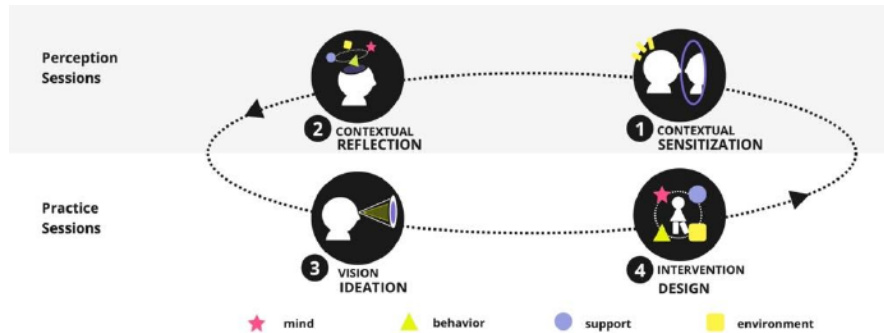


Figure 1: The Context4Change (C4C) framework consists of two sessions—Contextualization and Construction—and four steps: Contextual Sensitization, Contextual Reflection, Vision Ideation, and Intervention Design, the contextual reflection and intervention design session are guided by four key contextual factors: active mind, behavior, support, and environment.

Abstract

Physical inactivity (PIA) is pervasive in office environments, putting desk-bound workers at heightened risk of serious health issues. To help designers and researchers develop behavioral interventions that better align with end-user (desk-bound worker) contexts, we propose Context4Change (C4C), a participatory framework integrating behavioral and contextual design methods. C4C addresses shortcomings in existing top-down or technocratic approaches by emphasizing user experiences and contextual understanding. Through design workshops with diverse stakeholders (designers, researchers, and desk-bound workers), we identified key participant roles, interaction needs, and avenues for improving contextual communication. Our findings reveal how C4C broadens designers' perspectives, empowers users to articulate needs, and helps researchers translate contextual insights into actionable intervention objectives. In future work, we will refine C4C's processes

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to enhance its accessibility and conduct further multi-stakeholder studies to extend its impact on contextual intervention solutions for PIA in diverse workplace contexts.

CCS Concepts

• **Human-centered computing:** • **Interaction design;** • **Interaction design process and methods;**

Keywords

Participatory design, Office health Intervention, Contextual design, Behavior change, Physical Inactivity

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1 Introduction

“These office workers need more movement with technology kits,” propose a group of design students during a brainstorming session for a workplace health intervention with a PhD candidate as the facilitator. Yet none of them shares the day-to-day experience of rigid

deadlines, scattered break times, and cross-time-zone meetings that office employees face. Although they devise creative solutions, they remain uncertain whether these ideas fit the unpredictable contexts of end users' work environments. This scenario reflects a broader challenge in designing effective and contextually appropriate interventions for physical inactivity [14] in contemporary workplaces.

PIA is pervasive in today's offices, contributing to serious health issues such as obesity, cardiovascular diseases, and metabolic syndrome, along with increased mental health risks like anxiety and depression [23, 34]. Meanwhile, technological advancements and flexible work schedules have disrupted regular office routines, transforming them into dynamic "black boxes." Employees now juggle remote collaborations, spontaneous task reprioritizations, and fragmented break times, further complicating promoting physical activity (PA) and reducing sedentary behaviors (SB) [10]. These shifts underscore the need for design and research methods transition that accommodate rapidly changing office contexts.

Many existing behavioral interventions design approaches rely on theory-based models (e.g., Fogg Behavior Model [33], COM-B model [32]), structured design processes (e.g., Theoretical Domains Framework (TDF) [3], Intervention Mapping (IM) [4]), or persuasive technologies (e.g., nudging [31], gamification [28]). While these methods have produced successes, their top-down, technocrat orientations may overlook the diverse and evolving contexts of ever-changing workplace. Although academia offers abundant methods for cultivating empathy and gathering contextual insights, however rarely founded research that has considered the integration of contextual understanding and behavioral intervention design.

In this late-breaking work (LBW), we proposed Context4Change (C4C), a participatory framework that integrates behavior change theories with contextual design process [1]. With C4C, we aim at helping multi-stakeholder to co-create contextual behavioral interventions for tackling PIA in office. Therefore, our research question (RQ) is: How does C4C support multi-stakeholders to co design solutions for reducing office workers' PIA? To answer this RQ, we recruited 18 participants and conducted three rounds of design workshops to collect their feedback on C4C and observed how C4C was used to support their research and co-design process. Participants noted that C4C broadened designers' perspectives, helped end users articulate contextual needs, and provided pathways for researchers to translate contextual insights into design strategies. This LBW has made two main contributions: (1) Propose a participatory framework "C4C" that integrates behavior change theories with contextual design processes to support multi-stakeholder collaboration and contextual adaptation; (2) Evaluate C4C through workshops, demonstrating its potential to advance behavior change design in decentralized workplace environments. In future work, we will refine the C4C framework and explore its application in expert-led studies to uncover new opportunities for vitality toolkit design.

2 RELATED WORK

This section reviews prior research on **PIA intervention design in office settings**, discusses **behavior change theories and design frameworks**, and examines **design challenges in multi-stakeholder collaboration**.

2.1 Physical Inactivity Intervention Design in Office Settings

PIA among office workers is a global public health concern, linked to non-communicable diseases such as cardiovascular disease, metabolic syndrome, and mental health challenges [12]. In today's workplaces, contexts for PIA are diverse and dynamic, posing significant hurdles for designing effective interventions. Successful office-based behavioral interventions should address individual-level behavior change requirements while adapting to dynamic factors such as work stress, task prioritization, and collaboration demands. While some tools (e.g., smartwatches like Apple Watch [2] with sedentary reminders, sit-stand desks) demonstrate short-term efficacy, they often underperform during high-pressure periods or collaboration-intensive tasks [36]. This highlights the need for interventions that adapt to varying workplace contexts. Existing methods demonstrate varying levels of effectiveness in promoting physical activity. For example, Social Cognitive Theory (SCT) [26] enhances self-efficacy through behavior modeling and feedback mechanisms, making it widely used in behavior change interventions. However, SCT provides limited support for addressing the complexity of dynamic workplace contexts. Intervention designers should better understand these dynamics and translate contextual complexity into targeted intervention strategies in a sensible way [18, 24, 25].

2.2 Behavior Change Theories and Design Frameworks

Addressing PIA in dynamic office contexts requires theoretical support for integrating behavioral theories with design methods. While current frameworks offer varied support, rarely does a comprehensive method effectively address these challenges. The Participatory Intervention Model (PIM) [21] emphasizes user engagement, enhancing contextual understanding in the community, but lacks integration with behavior change theories. Conversely, the Behavior Change Wheel (BCW) [19] combining the COM-B model with intervention functions and policy categories, offers a solid theoretical foundation. However, while effective in health policy and educational programs, BCW provides limited support for dynamic contextual adaptation and user participation. IM and the TDF provide detailed tools for linking needs assessment with intervention strategies. IM's six-step process combines evidence with theory for systematic health intervention design [4]. TDF complements COM-B by providing in-depth behavioral analysis across multiple domains [6]. Yet, both lack support for dynamic user feedback and collaborative design. In HCI, persuasive technologies such as nudging and gamification enhance interventions through personalized prompts and motivational mechanisms, however, they are often limited by other contextual factors. While effective in fragmented office contexts, a method that fully addresses dynamic user participation and collaboration is needed.

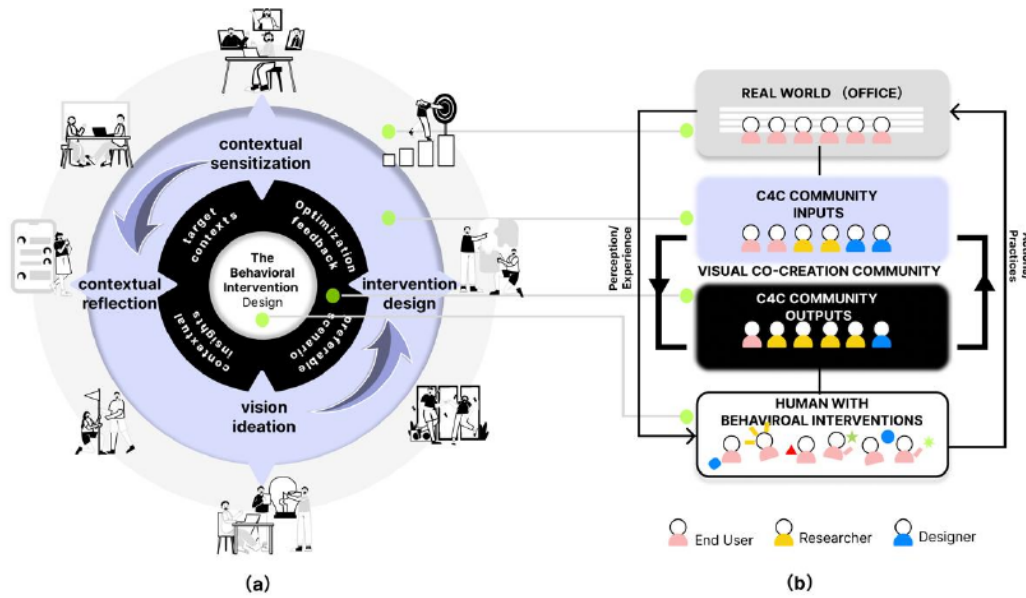


Figure 2: Relational Diagram: C4C's Operational Relationships and Collaborative Innovation approach: (a) This diagram depicts the interplay between C4C's operations, real-world office contexts, and contextual behavioral interventions. (b) It illustrates a scalable, participatory model generating contextual behavioral interventions, linking real-world office environments with virtual co-creation. The diagram highlights C4C's online community mechanisms in this study, fostering collective action and stakeholder collaboration (end users, researchers, designers) for workplace well-being.

2.3 Design Challenges in Multi-Stakeholder Collaboration

Developing office health interventions involves multi-stakeholder collaboration, where diverse goals and priorities must be reconciled. However, existing models often fail to bridge gaps among stakeholders with varying objectives—such as management's emphasis on productivity versus employees' focus on well-being [27]. Further complicating matters is the high degree of complexity in modern work settings [13]. Many established behavioral intervention design approaches overlook these dynamic aspects, resulting in interventions with suboptimal real-working context performance. More comprehensive and contextual design methods are therefore necessary to reconcile stakeholder priorities and effectively address complex, shifting workplace contexts.

3 THE C4C FRAMEWORK

To explore how behavior change models and structured intervention design processes facilitate stakeholder engagement in contextual inquiry and intervention co-design, we propose the C4C framework (see Figure 1). Inspired by contextual design processes and the COM-B model, which emphasizes the significance of context and has been widely validated in academia and practice, C4C comprises two main phases: Perception and Practice sessions. These sessions include four core steps (Contextual Sensitization, Contextual Reflection, Vision Ideation, and Intervention Design) and

four key contextual factors (mind, behavior, support, environment), supporting activities with behavior change theory.

How to Use: In Contextual Sensitization, participants collect data on user behaviors and environmental factors, recognizing the need for change and clarifying target contexts. This foundational work prepares for deeper exploration. Contextual Reflection applies COM-B's systematic perspective to examine motivation, support, environment, and behavior. (Terminology adjustments ensure accessibility for stakeholders from diverse backgrounds.) By identifying barriers and facilitators, this step highlights core contextual needs. Vision Ideation translates insights into future scenarios addressing user experience requirements, enabling stakeholders to integrate experiential and emotional considerations when envisioning desirable contexts. Finally, in Intervention Design, practitioners combine contextual visions and key factors to develop actionable behavioral intervention concepts, tested and refined through user feedback and prototyping to ensure alignment with user needs. We illustrated how the C4C approach iteratively interacts with office contexts and intervention design, while also showcasing stakeholders' collaboration dynamics. (see Figure 2).

The C4C framework stands out for its dynamic contextual adaptability, multi-stakeholder collaboration support, and effective translation of insights into practice. Compared to traditional behavior change frameworks like COM-B and BCW, C4C not only focuses on intervention functions and theoretical logic but also captures the dynamic needs of workplace environments through the "Contextual Sensitization" and "Contextual Reflection" phases, offering high

flexibility and adaptability. Furthermore, C4C emphasizes multi-stakeholder collaboration by defining clear roles, such as Reflective Explorers, Insight Seekers, and Creative Implementers, facilitating alignment between designers, researchers, and users, rather than relying solely on individual participation. Unlike frameworks such as TDF and IM, C4C bridges the gap between theoretical insights and actionable intervention strategies through specific activities such as preferable user experience collages during the “Vision Ideation” and “Intervention Design” phases. Overall, C4C provides a more flexible, practical, and participatory approach to address the complex contextual needs in workplace health intervention design.

4 METHOD

To evaluate the C4C framework and understand how diverse stakeholders perceive its core steps for contextual behavioral intervention design—while identifying typical participant roles and potential needs—we conducted a series of on-site/online workshops with 18 participants. These workshops employed generative techniques supported by digital and physical tools (e.g., Miro board [20], research probe kits, paper worksheets) to facilitate exploration, discussion, and application of the framework. Participants were divided into three groups, attending either in person or asynchronously online, with remote sessions enabled via videoconferencing to accommodate geographical constraints. All workshops were video and audio-recorded, and transcripts were produced for subsequent analysis. Ethical approval was obtained from the Ethics Review Board (ERB2023ID300) of Technology University of Eindhoven. All participants provided voluntary informed consent and received no compensation.

4.1 Participants

We recruited industrial designers, design researchers, and desk-bound office workers to represent key stakeholders in workplace PIA interventions—those designing, implementing, and experiencing behavior change—following participatory design principles [29]. Initially, 18 participants (12 females, 6 males), aged 25–38 ($M = 32.4$, $SD = 3.89$), all with Chinese backgrounds, were recruited. Due to scheduling conflicts, 3 withdrew, leaving a final sample of 15. This selection ensured representativeness across dynamic working contexts, promoting a shared understanding of evolving design method needs in workplaces with fluctuating schedules. Participants were assigned to one of three workshops (W1, W2, and W3) based on availability and backgrounds: W1 Design Researchers ($N=6$: 1m, 5f), W2 Industrial/Interaction Designers ($N=5$: 5f), and W3 Office Staff ($N=4$: 2m, 2f). Detailed demographic information is provided in Section 1 of the Supplementary Material.

4.2 Tools and Procedure

Drawing on the C4C framework, we developed a set of research probe toolkits and adapted them for both on-site and online workshops, providing a printable paper toolkit and an interactive online toolkit (using Miro Board). The toolkit details are available in Section 2 of the Supplementary Material. Unlike traditional participatory design, which relies on small-scale, face-to-face collaboration, our study explores a hybrid model where end users and designers engage asynchronously online, while researchers

participate synchronously in person. Inspired by social media engagement, this model allows office workers and designers to share insights and co-develop intervention strategies in a virtual space individually, storing community identity while ensuring privacy. The asynchronous format reduces social desirability bias, fostering open discussion and authentic experiences. It extends participation beyond geographical boundaries and enables a scalable, data-rich behavioral intervention design communication. Each workshop followed a structured approach: (1) Introduction, presenting the C4C framework focusing on PIA in office contexts; (2) Perception Sessions, where participants used the research probe toolkit to reflect on their office-based PIA; (3) Practice Sessions, envisioning preferable PA scenarios and designing behavioral interventions; (4) Survey Feedback, gathering qualitative reflections on experiences with the C4C framework. W1 lasted 2.5 hours in a university meeting room. W2 and W3 participants completed tasks asynchronously on Miro boards within one week.

4.3 Data Analysis

The analysis process followed the open coding [8] approach. Two researchers, and co-authors of this paper read through the transcribed data, independently marked relevant data to the research question and turned them into labeled statements. Next, the two researchers categorize the statements using affinity diagrams [17] to identify key themes. All the identified themes and clusters were reviewed, discussed, and revised by all the co-authors to validate the qualitative analysis.

5 RESULTS

Three key themes emerged from the affinity diagrams, namely: (1) engagement patterns and potential requirements; (2) key roles in the C4C framework; and (3) participants’ positive feedback alongside recommendations for improvement for the C4C framework. Participants are labeled as W#-P#. For example, W1-P1 represents Participant 1 from Workshop 1. The photos of the on-site workshop are available in Section 3 of the Supplementary Material.

5.1 C4C Application: Online Community Dynamics and Stakeholder Mindsets

The C4C framework’s application in online workshops unveiled distinct behavioral patterns and design/collaboration needs, influenced by online community dynamics and stakeholder mindsets. Design researchers (W1) excelled in the asynchronous, text-driven environment, systematically dissecting user contexts to map intervention possibilities. W1-P3 explained, “Structured online prompts enabled me to analyze user perspectives step-by-step, prioritizing strategic clarity over quick solutions.” Designers (W2), attuned to visual collaboration, actively engaged with shared Miro boards to prototype solutions but sought real-time user feedback, with W2-P4 stating, “The board sparked ideas, yet I need user critiques and more open, inspiring templates to iterate effectively.” Office workers (W3), initially unfamiliar with online tools for contextual reflection, adapted over time, marveling at C4C’s ability to uncover latent needs. W3-P4 noted, “The open-ended digital space let me express office challenges and plan actions at my own pace.” These patterns indicate W1 requires structured contextual analysis

guidance, W2 demands dynamic, interactive tools for co-design with users, and W3 thrives on adaptable, low-pressure platforms for self-expression, underscoring the need for low-learning-curve interactions in online collaboration.

5.2 Key Roles in the C4C Framework

Based on participant behavior and survey feedback, we identified distinct roles and mindsets causing misalignments among diverse participants using the C4C framework for contextual inquiry and intervention design. For instance, office workers might adopt designers' perspectives, neglecting their own needs, while designers might assume users' roles when defining contexts. We categorized three key roles with unique mindsets—Reflective Explorers, Insight Seekers, and Creative Implementers—to address these dynamics. Reflective Explorers focused on sensitization and reflection, deeply exploring contexts but rarely producing clear designs, as W3-P4 noted, "With clear motivations, I'll explore and experiment in work and life!" Insight Seekers valued theoretical depth but needed design guidance, with W1-P3 stating, "Contextual reflections and visions guided my concept, but I need tools to express ideas." Creative Implementers prioritized open-ended ideation, seeking novel triggers over combining existing solutions, as W2-P4 said, "I want tangible items or icons for interventions, not just existing solutions." These findings highlight diverse roles and mindsets, underscoring tailored support needs in C4C processes. However, such role misalignment phenomena may also occur in participatory design without the C4C framework, warranting further research to explore their prevalence and impact across design contexts.

5.3 Positive Feedback and Recommendations for Improvement

Participants praised the C4C framework's ability to expand design possibilities and stimulate innovative thinking, particularly in the reflection and visioning phases. W3-P2 commented, "I reflected on my daily behavior more deeply, discovering my core motivations," while W1-P3 added, "These four elements expanded possible design directions dramatically." However, participants also recommended improvements, such as providing more detailed guidance during the design phase, increasing flexibility to adapt the framework to diverse user contexts, and offering stage-specific feedback mechanisms. W1-P6 suggested, "The framework needs clearer guidance for transitioning from insights to intervention design. Specifically, contextual design is unfamiliar to me, some case studies would help clarify it." These insights highlight potential enhancements to improve the framework's adaptability and impact across varied design contexts.

6 DISCUSSION

In this section, we reflect on lessons learned, study limitations and future research opportunities.

6.1 Designing With Not For: Designers as Partners in Community-Driven Behavioral Intervention Design

Our findings reveal a shift in the role of designers in community-driven behavioral design, transitioning from traditional "solution providers" to being a design facilitator role of the co-creation group. This democratized process ensures final solutions are better aligned with diverse user needs and contexts. This transition aligns with the humanity-centered design principles in Design for a Better World [7], which advocate for designers as facilitators who co-create systemic solutions deeply rooted in users' environments. A similar example is highlighted in Dr. James Levine's book *Get Up!* [15] where a vitality team introduced the [16] concept to a company and provided guidance to their employees. Based on their schedules and preferences, employees independently designed action plans to reduce sedentary behavior, creating solutions tailored to their work contexts.

6.2 HCI Technology as a Communication and Co-Creation Bridge

This study highlights the transformative role of technology as a communication and co-creation bridge in HCI design practices. Tools like the Miro board enable asynchronous collaboration [11], allowing stakeholders to share ideas, exchange perspectives, and co-create designs without time or location constraints. Based on the user contextual feedback and the visualized data such as future vision emoji collages stored in the online tools, we redesigned four intervention concepts targeting the four prolonged sedentary contexts in office settings. For details, refer to Section 4 in Supplementary Material. By streamlining the exchange of contextual insights, these tools transition from static utilities to mechanisms for dynamic engagement and iterative refinement, supporting continuous improvement and responsiveness to user feedback [5]. This aligns with Mediation Theory [35], which highlights how technology mediates human experiences and actions, shaping interactions and fostering meaningful engagement. Technological tools become active participants in the design process, bridging gaps between stakeholders and ensuring interventions are both contextually adaptive and user centered.

6.3 Expanding the Scope of C4C Application

This study engaged designers and researchers using C4C to design vitality solutions for office workers, but its potential extends further. We envision C4C integrating into employee care strategies across diverse organizations—small businesses, corporations, or nonprofits—with managers and HR as key collaborators, applicable to remote, hybrid, and traditional office settings. The pandemic has intensified issues like prolonged virtual meetings, reduced social connections, and increased sedentary behavior in remote work settings [22], necessitating a rethinking of well-being and workplace dynamics. C4C's online asynchronous mode can better support remote team members in co-designing health interventions, leveraging its flexible structure to foster collaboration across distances. For example, Welke [30] at Adidas highlighted that while employees are responsible for healthy habits, companies should foster active

workplaces. Similarly, Google [9] designs playful offices to ensure happy, healthy employees, believing this leads to organizational success. These cases align with our vision of C4C as a platform for stakeholders (e.g., employees, researchers, architects, HR managers) to co-design vitality ecosystems, not just isolated fixes. By adopting C4C, organizations of varying sizes and structures can address immediate well-being challenges and build sustainable health and productivity frameworks, particularly in navigating remote and hybrid models. This adaptability enhances C4C's universality across evolving work contexts.

6.4 Limitations

We are aware of the limitations of this study. First, the asynchronous format posed challenges, as there was no direct communication between organizers and online participants, which could lead to misunderstandings of rules and concepts or missed insights generated through conversations. Second, some online workshops required participants to complete tasks independently, missing the opportunity to explore collaborative interactions in a multi-person online setting. Third, while the small and diverse sample size provided valuable insights, its generalizability is limited. Additionally, this study focused on a specific subset of office stakeholders, which may further restrict the applicability of the findings. These limitations will be addressed in future work by refining the workshop format and expanding participant recruitment, while also including decision-makers, HR specialists, and health professionals to validate C4C's adaptability. As a scalable model, C4C could be further tested in hybrid work settings and cross-organizational collaborations.

7 CONCLUSION

This study introduced the C4C framework to support designers and stakeholders in collaboratively addressing the contextual complexities of workplace behavioral interventions. By integrating behavior change theories (COM-B) with contextual design processes, the framework enables user participation across four steps: Contextual Sensitization, Contextual Reflection, Vision Ideation, and Intervention Design-to create adaptive, context-sensitive solutions. Workshops with design researchers, designers, and office workers demonstrated the framework's effectiveness in uncovering contextual needs and aligning interventions with real-world applications. Participants praised its ability to foster empathy, collaborative design, and broader perspectives but suggested improvements in practical guidance and flexibility for diverse contexts. Future work will refine the framework, incorporate evaluative mechanisms, and test its application in organizational settings, promoting awareness of employee well-being and enabling the co-creation of impactful workplace vitality solutions.

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